

Late blight: pathogen variability and disease resistance breeding in Ecuador



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PREVIEW

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Ricardo Delgado

Thesis

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PREVIEW

Dedication

To my wife Jennifer and my daughter Silvia Pilar

To my parents Julio César and Pilar

PREVIEW

CHAPTER 1

General Introduction

Potato crop

The potato belongs to the Solanaceae family which was first domesticated in South America in the surroundings of Lake Titicaca which today is the border between Perú and Bolivia about 8000 years ago (De Jong, 2016). It allowed the development of societies across the Andes, including the Inca Empire by the time of the arrival of the Spanish conquistadors. After this, potatoes were initially introduced to the Canary Islands and in Seville, Spain in the 1500s and, from there they were later distributed to the rest of the world (Brown & Henfling, 2014; Bentley, 2015, De Jong, 2016). Potato tubers are widely consumed due to their high carbohydrate and protein contents, additionally they are a valuable source of nutrients like Potassium, Calcium, Phosphorus, Copper, Magnesium and vitamins C and B (Camire et al, 2009; Stoorey 2007, Navarre et al, 2014).

Currently, the potato is the fifth most important crop in the world after, corn, wheat, rice and sugar cane with 376 million tons produced, it occupied the seventeenth place for cultivated area with 19 million ha in 2016 (<http://www.fao.org/faostat/en/#home>). It is cultivated around the world, with China, India, Rusia and Ukraine being the top four countries in area harvested and production, whereas The United States of America, New Zealand, Germany, Denmark and The Netherlands have the highest yields per ha (Table 1). In South America, the top producers are Peru, Brasil, Colombia and Argentina, meanwhile the highest yields are obtained in Brazil, Argentina, Chile and Uruguay (Table 2).

Potato in Ecuador

One of the most important crops in Ecuador is the potato, therefore it is an important part of the dietary consumption, especially in the highlands (Devaux *et al.*, 2010). The annual consumption per capita in the country is about 31.8 kg per year (Devaux *et al.*, 2010). It is cultivated in the Ecuadorian highlands from 2.000 to more than 3.600 meters above sea level (masl). Small scale farmers mainly cultivate it (around 45.000 of them) in an area of 29.635 hectares in 2016 (INEC, 2016) (Table 3). Most of the small farmers, about 88%, cultivate their potatoes in farms of no more than 20 ha (Rivadeneira, 2009). Even more, about 50% of the growers produce their potatoes in properties of less than 1 ha (Monteros-Guerrero, 2016). Potatoes are mainly produced in the provinces of Chimborazo (7.450 ha), Carchi (7.241 ha), Cotopaxi (3.909 ha), Pichincha (2.969 ha), Tungurahua (2.077 ha), and Cañar (1.995 ha) (Table 3 & Figure 1). In 2016, 422.589 tons were harvested (INEC, 2017). The main producing provinces are Carchi which represents 36.14 % of the total harvest, followed by Chimborazo (26.88%) and Cotopaxi (11.11 %) (Table 3). The national average yield was 12.71 t/ha in 2016, with the highest yields obtained in Carchi province (Table 3).

Ninety per cent of the potatoes are consumed fresh and the rest are used as chips and french fries (Devaux *et al.*, 2013). In the Northern provinces (Carchi and Imbabura) the growers prefer the Superchola variety which has a pink skin and yellow flesh. In the central provinces (Pichincha, Cotopaxi, Tungurahua, Chimborazo and Bolivar) the preferred variety is INIAP-Fripapa. In the southern provinces (Cañar, Azuay and Loja), Bolona variety, which is a landrace has the preference of the farmers; its tubers have a creamy coloured skin with creamy coloured flesh and a round shape (Devaux *et al.*, 2013) (Figure 2). According to Monteros-Guerrero (2016), Superchola variety (55%) is the most planted potato variety in the country followed by INIAP-Fripapa (5%) and in lower percentages several other clones.

Other particularities about the potato cultivation in Ecuador include that about 85% of the growers use their own seed obtained from the previous harvest and not from formal seed sources. Additionally, due to their small scale, most of them have difficulties with using technologies like irrigation, resulting in just 33% of the farmers with access to it (Monteros-Guerrero, 2016). Regarding the educational level of the farmers, 58% had elementary school partial or completed, 26% high school, 4% university and 7% no formal education (Flores *et al.*, 2012). Aside from all this, there are biological stresses that affect the crop, reducing the yields and among them late blight had been consistently the most important one (Oyarzún *et al.*, 2002; Devaux *et al.*, 2010; Monteros-Guerrero, 2016; Navarrete *et al.*, 2017).

Table 1. Potato statistics of main countries on area harvested (ha), production (t) and average yield (t/ha) in 2016.

Country	Area harvest (ha)	Country	Production (t)	Country	Average Yield (t/ha)
China	5812865	China	99065724	United States of America	49.02
India	2130000	India	43770000	New Zealand	48.99
Rusia	2030858	Rusia	31107797	Germany	44.42
Ukraine	1311600	Ukraine	21750290	Denmark	42.48
Bangladesh	475699	United States of America	19990950	Netherlands	42.00
United States of America	407810	Germany	10772100	Australia	40.41
Canada	342409	Bangladesh	9474099	Jordan	40.05
Nigeria	333100	Poland	8872445	Ireland	39.11
Poland	311620	France	6834680	France	39.01
Peru	310698	Netherlands	6534338	United Kingdom	38.65
World	19246462	World	376826967	World	19.58

FAOSTAT (<http://www.fao.org/faostat/en/#home>).

Table 2. Potato statistics of South American countries on area harvested (ha), production (t) and average yield (t/ha) in 2016.

Area	Area harvested (ha)	Production (t)	Yield (t/ha)
Argentina	59834	1750000	29.25
Bolivia	181708	1073744	5.91
Brazil	129842	3851396	29.66
Chile	53485	1166024	21.80
Colombia	121920	2354862	19.31
Ecuador	29635	422589	14.26
Paraguay	250	3625	14.50
Peru	310698	4400295	14.16
Uruguay	4424	89000	20.12
Venezuela	20202	391433	19.38

FAOSTAT (<http://www.fao.org/faostat/en/#home>).

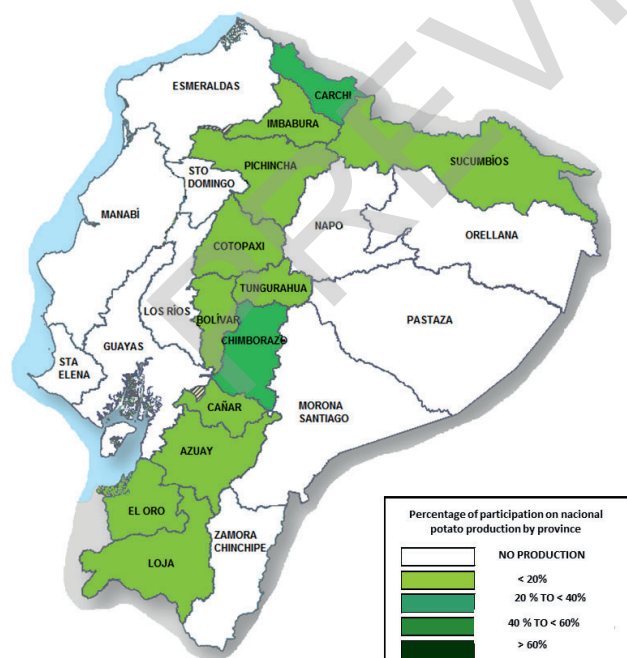


Figure 1. Potato production areas and percentage by province in Ecuador (INEC, 2017).

Table 3. Area harvested (ha) and pro-duction (tons) in the provinces where potatoes are cultivated in Ecuador.

Province	Area harvested (ha)	Percentage (%)	Production (t)	Percentage (%)	Average Yield (t/ha)
Azuay	1341	4.53	7298	1.73	5.44
Bolívar	1611	5.44	12262	2.90	7.61
Cañar	1995	6.73	8461	2.00	4.24
Carchi	7241	24.44	152742	36.14	21.09
Cotopaxi	3909	13.19	46929	11.11	15.25
Chimborazo	7450	25.14	113588	26.88	12.01
Imbabura	923	3.11	9625	2.28	10.43
Loja	32	0.11	134	0.03	14.21
Pichincha	2969	10.02	42203	9.99	13.12
Tungurahua	2077	7.01	27257	6.45	18.69
El Oro	10	0.03	32	0.01	3.18
Sucumbíos	75	0.25	2057	0.49	27.31
Total	29635	100	422589	100	-
Average	-	-	-	-	12.71

/Adapted from data from INEC (2017).



Figure 2. Tubers from Superchola (A), INIAP-Fripapa (B) and Bolona (C) potato varieties (Pumisacho & Velazquez, 2009).

Late Blight around the world and in Ecuador

Late blight, called in Ecuador ‘tizón tardío’ in Spanish or ‘lancha’ in kichwa, is caused by the oomycete *Phytophthora infestans* (Mont.) de Bary. This is the main disease of potatoes worldwide (Thurstun & Schultz, 1981). It was the cause of the known ‘Irish famine’ which caused the death of a million people and the immigration of another million in the 1840s (Nowicki et al, 2012; De Jong, 2016). The disease is still a matter of concern worldwide due

to the epidemics that reduce the yields and incomes among others because of the increase of the production costs in order to control this problem (Fry, 2008). The European Union has estimated a loss of 1 billion euros annually due to the disease (Haverkort et al, 2008). Also in the United States, the costs for the fungicides applied for the control of late blight have been calculated to be around 300 to 500 dollars per hectare (Johnson et al, 2000; Guenther et al, 2001).

In Ecuador, late blight is the main constraint of this crop in the country (Oyarzún *et al.*, 2002). The first time the pathogen was described in Ecuador it infected the fruit bearing crops, sweet cucumber (*Solanum muricatum*) and tzimbaló (*S. caripense*) (Lagerheim, 1890). Years later, it was observed causing blight in potatoes (Pachano, 1918). Epidemics caused by this pathogen led to losses that can range from 28 to 100 % of the yield varying due to host resistance and weather conditions (Morales *et al.*, 1995).

The pathogen can infect leaves, stems, petioles and tubers (Oyarzún *et al.*, 2002). Leaf symptoms consist of brown spots of necrotic tissue often surrounded by a pale green to yellow halo (Figure 3). Lesions can expand and coalesce with each other, killing the leaflets, leaves and eventually the whole plant (Figure 3) (Thurston & Schultz, 1981). When high moist weather conditions occur sporangia are formed in the abaxial leaf surface (Figure 3) (Thurston & Schultz, 1981). Stem lesions can also occur as dark brown spots which can break by wind or other movements (Figure 3) (Perez & Forbes, 2010). The foliage damage affects the photosynthesis reducing the number and weight of the tubers. The damage can be so extensive that the whole plant is destroyed (Thurston & Schultz, 1981).

The life cycle of the pathogen consists of asexual reproduction through sporangia which germinate directly producing mycelia or by realizing motile zoospores (Figure 4 & Figure 5). These spores later lose their flagella and germinate causing infection through a germ tube. Until this moment, only the asexual cycle of the pathogen seems to exist in Ecuador (Oliva *et al.*, 2002). The disease cycle starts from the sporangia grown on the surface of the infected tissue which has gotten necrotic. These can be dispersed by rain and wind. Once landed on susceptible tissue, sporangia can germinate directly producing mycelia which infect the plant or by realizing zoospores. The pathogen can survive in infected tubers, debris and volunteer plants. Infection of tubers can occur but is rarely observed in Ecuador (Oyarzún *et al.*, 2005, Kromman *et al.*, 2008b), possibly due to suppressiveness of the soil, either by biological and/or soil factors (Oyarzún *et al.*, 2011, Orquera-Tornakian *et al.*, 2018). Late blight epidemics could start soon after the emergence through infected sprouts from zoospores or sporangia from infected potato plants in the field (Kromman *et al.*, 2008b).

Two clonal lineages, EC-1 and US-1, both of the A1 mating type have been identified in Ecuador (Forbes *et al.*, 1997). The EC-1 lineage, is predominant in the potatoes and US-1 in tomatoes (Forbes *et al.*, 1997; Oyarzún *et al.*, 1998). These populations have been changing through the years, from a predominance of avirulent isolates (INIAP, 1976) to complex races (Forbes *et al.*, 1997, Tello, 2008). The EC-1 lineage was found to be more aggressive than the US-1 (Ordoñez *et al.*, 1998).

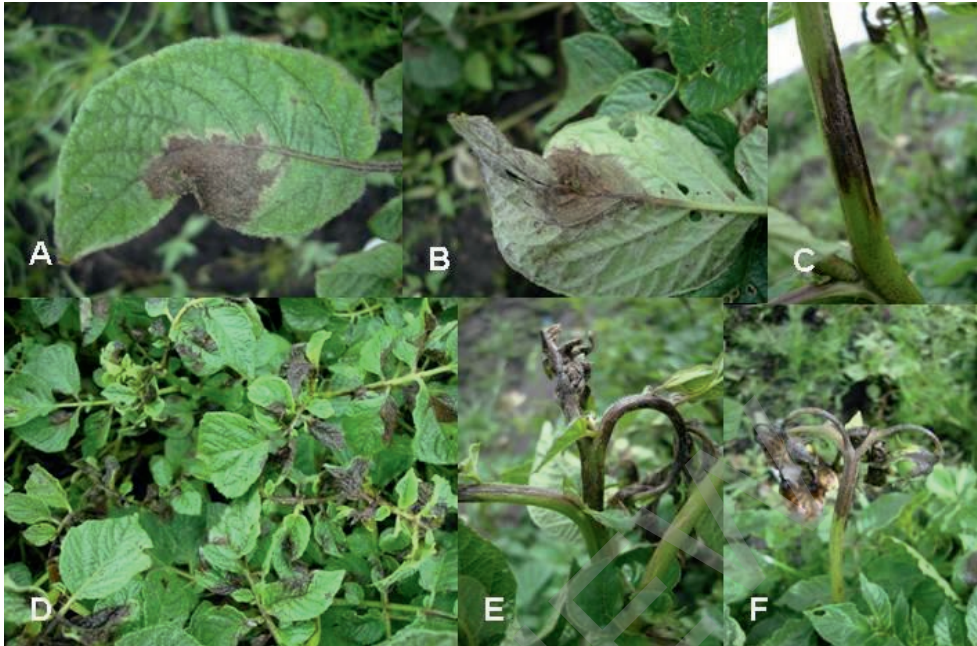


Figure 3. Symptoms of late blight in potato: Lesion on a potato leaflet caused by *P. infestans* surrounded with a yellow halo (A); Sporulation on the abaxial surface of leaflet (B); Stem lesion (C); Necrotic lesions on foliage (D); Blight lesions on the apex of the plant (E); Blight of the inflorescence (F).

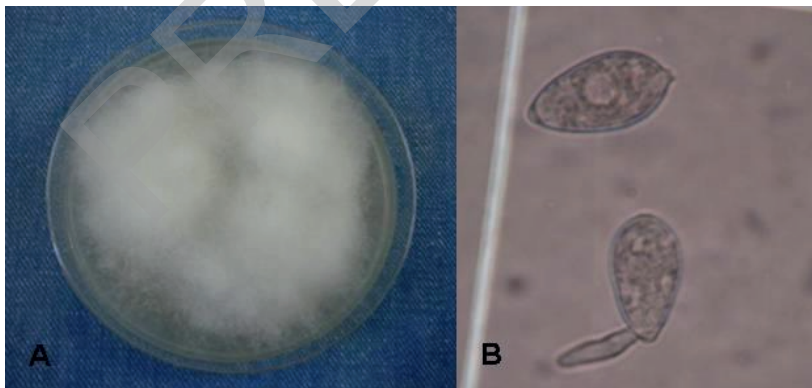
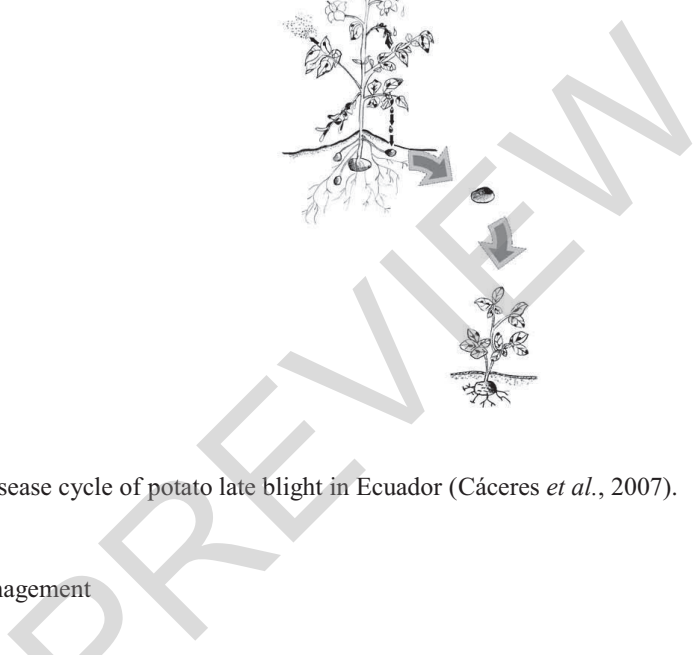


Figure 4. Colony of *P. infestans* growing in Rye Agar media (A); Sporangia of *P. infestans* (B).



Disease management

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as reported by Crissman *et al.*, (1994), Cole *et al.*, (1997) and Yanggen *et al.*, (2003) who reported cases with symptoms of skin irritation.

The other control method available is the planting of late blight resistant varieties. This is thought to be the most suitable control because it is supposed to be easy to adopt. Despite this, it has been observed in different countries that late blight susceptible varieties are preferred by farmers due to their quality and consumer preferences, like is the case for the Granola and Superchola varieties in Indonesia and Ecuador, respectively (Forbes, 2012). Other advantages of the use of resistant varieties are that they are friendly for the environment and safe for farmers.

The main objective of breeding potatoes in Ecuador is obtaining improved varieties with valuable traits including late blight resistance. For this purpose different sources of genes are used, among them potato landraces which include species like *S. phureja*, *S. andigena* and *S. chaucha* (Monteros-Altamirano, 2011). Additionally introduced germplasm from other locations is used for obtaining improved varieties. The breeding process until the release of a new variety takes many years (10 -15) (Cuesta, 2011; Cuesta et al, 2015). The obtaining of a new potato variety has to include several traits apart from late blight resistance, such as earliness and nutritional contents among others. Diverse factors may affect the breeding for late blight resistance, such as pathogen population and environment. In the case of pathogen population, it represents a complication for the breeder since it has been observed that late blight resistant varieties, were quickly overcome by the pathogen and thus the varieties status changed from resistant to susceptible relatively soon after they were released, as observed with Arka (Tanzania), Capiro (Colombia), INIAP-Cecilia, INIAP-Gabriela, Superchola (Ecuador), Canchan and Amarilis (Perú) (Forbes 2012). This leads to the constant need for new sources of resistance to be included in the breeding plan.

The development of the technologies for the potato crop in Ecuador, which includes improved varieties, integrated pest and crop management are the responsibility of the National Agricultural Institute of Ecuador (Instituto Nacional de Investigaciones Agropecuarias – INIAP), with emphasis on applied research.

Scope of this thesis

Ecuador has as a valuable resource in the potato landraces which have been grown for many decades by the farmers. They may be a source of quantitative resistance to late blight since they have survived the epidemics of late blight induced by a new aggressive population in the country. The objective of this thesis is to characterize the resistance of Ecuadorian potato varieties, understand why a clonal population of *P. infestans* makes it difficult to obtain durable resistance, and attempt to identify QTLs associated to the resistance against this important and devastating disease.

In chapter 2, the potato breeding work performed in Ecuador is presented and analyzed. Methodologies, germplasm utilized, and main achievements are presented. Perspectives for future development are discussed.

In chapter 3, the diversity of the populations of *P. infestans* associated to potato landraces is studied. The pathogenic diversity in the country is analyzed and the genetic variability is evidenced.

In chapter 4, a screening of late blight resistance is performed with a group of selected landraces, not reported elsewhere, in Carchi province with the aim to characterize their reaction against the disease and identify the possibilities of incorporating some of them as new parental clones for breeding.

In chapter 5, an evaluation of late blight field resistance among a selected group of landraces, collected recently, is performed and an analysis of the perception of this disease by small farmers is made. Their value for breeding purposes and reasons for their use and existence until the present days are discussed.

In chapter 6, the behavior of a group of selected potato varieties is studied in different sites in Chimborazo province. The genotype x environment interaction is analyzed and discussed.

In chapter 7, a QTL analysis is performed in a segregating population of a cross between two *S. phureja* landraces which were shown to carry some level of quantitative resistance against late blight in an attempt to see whether QTL could be identified and if yes whether these could be combined to obtain a higher level of resistance.

In chapter 8, a discussion on the main findings of this thesis is presented. Recommendations for future research are made.

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Chapter 2

Breeding for potato late blight resistance in Ecuador: The National Agricultural Research Institute's INIAP's efforts

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Abstract

Potato is one of the most important food crops in Ecuador. Late blight is the main disease that affects potatoes in the country. Breeding efforts for obtaining improved potato varieties have been carried out by the Instituto Nacional Autónomo de Investigaciones Agropecuarias (INIAP) soon after its creation in the late 1950ties. The objective of this assay is to review the breeding for potato late blight resistance in Ecuador carried out by the INIAP, the strategies followed and results achieved in the last 60 years.

Keywords: *Phytophthora infestans*, *Solanaceae*, Oomycete

Introduction

Potato is one of the most important staple crops in Ecuador. It is an important part of the diet of all people in the country, especially in the highlands (Devaux *et al.*, 2010). Potatoes are cultivated in the Ecuadorian highlands (Figure 1), mostly by small scale farmers (Andrade *et al.*, 2002). Potatoes are grown all year around mainly based on the seasonal rainfall and the access to irrigation. The plantings are made from May to June and October to December. The average national yield was in 2016, 12.21 t/ha (INEC, 2017), which is lower than those obtained in Colombia (19.31 t/ha) and Perú (14.16 t/ha) (<http://www.fao.org/faostat/en/#home>). This may be explained by the fact that most of the farmers use seed tubers selected from the harvested production (Flores *et al.*, 2013), which is true for about 85% of the seeds used in the country (Monteros-Guerrero, 2016). This situation leads to a diminished sanitary quality of the seeds necessary to ensure good yields (Navarrete *et al.*, 2017).

Among the factors that limit the yield, there is late blight which is caused by the oomycete *Phytophthora infestans* (Mont.) de Bary and it is the main disease of potatoes in the country (Oyarzún *et al.*, 2002). The pathogen was first reported in Ecuador in 1890 infecting *Solanum muricatum* and *S. caripense*, probably the first report of a plant disease in the country (Lagerheim, 1890). Later, it was confirmed to cause blight in potatoes as well (Pachano, 1918). It can induce severe losses depending on host susceptibility and weather conditions that might favor the epidemics (Morales *et al.*, 1995). For the management of the disease, two alternatives have been applied; chemical control by the application of fungicides and the use of resistant varieties. Chemical control is used in the crop in order to prevent epidemics in commercial fields (Oyarzún *et al.*, 2002). It may require repeated sprays over the plants; they can be as high as 14 times as has been observed in Carchi province (Yáñez *et al.*, 2008). Another problem due to the high use of the fungicides is that they affect the health of the farmers as reported by Crissman *et al.*, (1994). The most suitable control measure is the use of resistant varieties because this does not require the use of fungicides anymore and it is supposed to be easy to adopt. The objective of this essay is to review the breeding for potato late blight resistance in Ecuador in the last 60 years, the strategies followed, results achieved and future perspectives.

Potato Breeding in Ecuador

The breeding efforts in the country, started in Ambato in the 1910's with the trials set by the Quinta Normal de Agricultura- an agricultural school- where native potatoes and newly introduced varieties were evaluated for late blight resistance (Pachano, 1918). Some years later (in the 1930's), crosses among *S. andigena* local varieties were performed by Manuel Bastidas, a private breeder from the Carchi province producing some improved varieties which were planted in that location (Estrada, 2000). The work of this breeder was followed by his son Germán Bastidas who released some varieties including Superchola (Bastidas, 1991), which is still one of the most cultivated commercial varieties in Ecuador (Pumisacho & Velasquez, 2009; Monteros-Guerrero, 2016).

The National Agricultural Research Institute of Ecuador (INIAP) was created in 1959. In 1961, a collection of three hundred clones was received from the Universidad Central del Ecuador and transferred to the Potato Program. This germplasm was named as Colección Ecuatoriana de la Papa- CEP (Ecuadorian Potato Collection) starting in this way the potato breeding activities of INIAP (Albornoz & Morillo, 1968).

Breeding Objectives of INIAP's Potato Program

Nowadays, the potato program has as breeding objectives to obtain improved varieties adapted to the conditions of the country, with the following attributes: high yielding (>30 t/ha), earliness (growth cycle shorter than 150 days), resistant against main biotic stresses (*Phytophthora infestans*, *Rhizoctonia solani*, *Pectobacterium* sp., *Globodera* sp. and viruses), resistant against abiotic stresses (drought, high or low temperatures, poor soils), suitable for fresh consumption (round shape, high contents of iron and zinc, and antioxidants) or industrial purposes like chips and fries (round to oblong shape, high dry matter content, etc) (Figure 3) (Cuesta, 2011; Cuesta *et al.*, 2015). Breeding of varieties includes different

approaches like clonal selection and hybridization which can take more than 10 years (Figure 2, 4).

Breeding strategies

Initially the breeding efforts consisted of the selection of clones from local germplasm and introduced from other origins like Colombia. Hence, in 1965, INIAP-Santa Catalina, a late blight resistant variety was released. This was the first variety offered by INIAP (Albornoz & Ortuño, 1968).

In collaboration with the International Potato Center (Centro Internacional de la Papa -CIP) in Peru, advanced clones were introduced and varieties were selected. Most of them had monogenic resistance which was effective when the varieties were released, but quickly were overcome by the pathogen (Revelo *et al.*, 1997). This was due to a shift of the dominant population of *P. infestans*, which was more aggressive (Forbes *et al.*, 1997) and has a high genetic diversity (Chapter 3; Delgado *et al.*, 2013).

During part of the 1990's a strategy for breeding based on quantitative resistance against late blight was implemented. The aim was to obtain varieties with durable resistance. It consisted of the screening of clones from the CEP and other origins for late blight resistance at field level. Additionally, the clones were tested using a detached leaf assay in the laboratory with a complex race and a race 0 of *P. infestans*. The purpose was to select progenitors that had resistance but no major genes, thus selecting for quantitative or so called field resistance against late blight (Cuesta *et al.*, 1999). To this scheme a recurrent selection procedure was added so that after some cycles of field trials advanced clones were selected as progenitors and introduced into the breeding process by crossing them with other parental lines and starting over the selection on the resulting materials.

A new approach was introduced in the Potato Program of INIAP, the so called Participatory breeding (PB). The purpose was to improve the breeding process by the inclusion of the criteria of the farmers, consumers, and the industry. All these had the goal to improve the acceptance and adoption of the new varieties. The PB started in 1992 as part of the breeding scheme in INIAP's Potato Program (Capelo, 1998). The procedure included the evaluation of different traits by selected groups of potential users of the varieties, such as farmers, consumers, industrial representatives among other actors involved in the potato chain. The traits under evaluation were late blight resistance, yield and organoleptic characteristics (Capelo, 1998, Cardenas *et al.*, 1998). Varieties such as INIAP-Raymipapa and INIAP-Suprema were selected using this approach (Cuesta *et al.*, 2000). PB is nowadays a routine part of the breeding process carried out by INIAP's Potato Program.

Other strategies have been explored over the years as well, like mutation breeding, with some results. Chacon & Forbes (2000) from CIP, irradiated plants of the varieties INIAP-Esperanza and INIAP-Cecilia. They observed that some of the plants were more resistant to late blight than the non-irradiated controls. They did not find immunity in the individuals so it was concluded that the irradiation induced variation in quantitative resistance. Years later, Lopez & Yáñez (2010) from INIAP, tried to induce mutations by irradiation in the late blight susceptible variety Superchola. Eighteen mutants performed better than untreated Superchola,